

Stat 3113: ANOVA & Factorial Design

Key Definitions

Factor: Independent variable (categorical) being manipulated.

Level: Specific values/categories of a factor.

Response: Dependent variable (numerical) measured.

Balanced Design: Equal sample sizes in all groups.

Interaction: Effect of one factor depends on level of another. Non-parallel lines.

Between-Group Variation: Variation due to treatment/factor effects.

Within-Group Variation: Variation due to random error/noise.

ANOVA Fundamentals

Purpose: Compare means of $k \geq 3$ groups.

Why not t-tests? Multiple t-tests inflate Type I error. ANOVA controls experimentwise error rate.

Hypotheses:

- $H_0 : \mu_1 = \mu_2 = \dots = \mu_k$ (All means equal)
- H_a : At least one μ_i differs

F-Ratio Logic: $F = \frac{\text{Signal}}{\text{Noise}} = \frac{\text{Between Variance}}{\text{Within Variance}}$

- Always $F \geq 0$
- $F \approx 1 \implies$ No effect (H_0 True)
- Large $F \implies$ Significant difference

One-Way ANOVA

Notation: N = Total subjects, k = Groups

Degrees of Freedom:

- $df_{Total} = N - 1$
- $df_{Between} = k - 1$
- $df_{Within} = N - k$
- Check: $df_{Tot} = df_{Bet} + df_{Wit}$

Sums of Squares: $SS_{Total} = SS_{Between} + SS_{Within}$

Mean Squares: (NOT additive)

- $MSTr = SS_{Between} / df_{Between}$
- $MSE = SS_{Within} / df_{Within}$

Test Statistic: $F = \frac{MSTr}{MSE}$

Assumptions

1. **Independence:** Random sampling
2. **Normality:** Normal Q-Q plot of residuals linear
3. **Equal Variance:** Residuals vs Predicted uniform band (no megaphone)

Two-Way ANOVA

Sources: Factor A, Factor B, $A \times B$, Error

Degrees of Freedom:

- $df_A = (\text{Levels of A}) - 1$
- $df_B = (\text{Levels of B}) - 1$
- $df_{A \times B} = df_A \times df_B$
- $df_{Error} = df_{Total} - (df_A + df_B + df_{A \times B})$

Main Effects: Effect of one factor averaged over levels of the other.

Analysis Hierarchy (Stop Rule)

Test highest-order interaction FIRST.

- **If Interaction Significant ($p < \alpha$):**
 - **STOP.** Do NOT interpret main effects.
 - Conclusion: Factors interact.
- **If NOT Significant ($p > \alpha$):**
 - Proceed to test Main Effects.

Factorial Experiments

Treatment Combinations: Product of levels

- Ex: 3 factors, 2 levels each $\rightarrow 2^3 = 8$ runs
- Ex: Levels 3, 5, 2 $\rightarrow 3 \times 5 \times 2 = 30$ runs

Model Adequacy: "Whole Model" p-value.

If $p < \alpha$, model is significant.

Post-Hoc: Tukey's HSD

When: (1) Rejected H_0 , (2) $k \geq 3$ groups

Connecting Letters Report:

- Levels sharing letter: NOT significantly different
- Levels NOT sharing letter: ARE different

Inference Recap

Confidence Interval: Estimate $\pm$ Margin of Error. "95% confidence" = method reliability. μ_0 outside CI $\iff$ Reject H_0 .

Significance Level (α): Threshold for rejecting H_0 (usually 0.05).

p-value: Probability of data this extreme if H_0 true.

Type I Error (α): Reject true H_0 (False Positive)

Type II Error (β): Fail to reject false H_0 (False Negative)

Power ($1 - \beta$): Probability of detecting real effect

Reading JMP Output

ANOVA Table:

- **Prob > F:** p-value
- If p-value < 0.05: Reject H_0
- **Effect Tests (Factorial):**
 - Check interaction term p-value first
- **Tukey's HSD:**
 - Connecting Letters Report
 - Ordered Differences (p-values < 0.05)

Stat 3113: Regression & Correlation

Key Definitions

Response Variable (Y): What we predict (Dependent).

Explanatory Variable (X): Used to predict (Independent/Predictor).

Residual (e): $e = \text{Observed} - \text{Predicted} = y - \hat{y}$

Least Squares Line: Minimizes $SSE = \sum (y_i - \hat{y}_i)^2$

Slope (β_1): Rate of change. If $\beta_1 = 0$, no linear relationship.

Intercept (β_0): Value of Y when $X = 0$. Only interpret if $X = 0$ is in data range.

Extrapolation: Predicting outside data range. Unreliable.

Influential Point: Outlier in X that drastically changes slope.

Lurking Variable: Hidden variable causing correlation (e.g., weather $\rightarrow$ ice cream sales AND shark attacks).

Correlation (r)

Pearson's Correlation:

- Measures strength & direction of LINEAR relationship
- Range: $-1 \leq r \leq 1$
- Unitless, Symmetric ($r_{xy} = r_{yx}$)
- $r > 0$: Positive (uphill)
- $r < 0$: Negative (downhill)
- $r \approx 0$: No linear association

Warning:

- NOT resistant to outliers
- **Correlation $\neq$ Causation**

Scatterplots

Axes: Explanatory (X) horizontal, Response (Y) vertical

4 Aspects to Describe:

1. **Form:** Linear, Curved, Random
2. **Direction:** Positive, Negative
3. **Strength:** Tight vs loose
4. **Outliers/Clusters**

Simple Linear Regression

Model: $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$

Fitted Line: $\hat{y} = b_0 + b_1 x$

- $\hat{y}$: Predicted value

- b_0 : Intercept estimate

- b_1 : Slope estimate

Slope Interpretation: "For every 1-unit increase in X , Y changes by b_1 ."

Intercept: "Predicted Y when $X=0$." (Only if $X=0$ is reasonable)

Inference for Regression

Hypotheses:

- $H_0 : \beta_1 = 0$ (No linear relationship)

- $H_a : \beta_1 \neq 0$ (Linear relationship exists)

Testing:

- t-test on slope: "Prob t —t—"

- F-test: "Prob F " (same for SLR)

- If $p < \alpha$: Reject H_0 , significant relationship

Model Fit: R^2

Coefficient of Determination:

- Formula: $R^2 = (r)^2$ (for SLR)

- Range: $0 \leq R^2 \leq 1$

- **Interpretation:** "% of variation in Y explained by linear relationship with X "

- Higher R^2 = Better fit

Assumption Checking

1. Linearity

- **Check:** Scatterplot or Residuals vs X

- **Good:** Random scatter around 0

- **Bad:** Curved pattern

2. Constant Variance

- **Check:** Residuals vs Predicted

- **Good:** Uniform band

- **Bad:** Megaphone/funnel

3. Normality

- **Check:** Normal Q-Q Plot of Residuals

- **Good:** Points follow line

Common Pitfalls

- **Extrapolation:** Don't predict outside data range
- **Causation:** Correlation doesn't prove it
- **Intercept:** Only meaningful if $X=0$ is reasonable
- **Influential Points:** Can dramatically alter slope

JMP Regression Output

Parameter Estimates:

- **Intercept Estimate:** b_0

- **X Variable Estimate:** b_1 (Slope)

- **Prob t —t—:** p-value for that term

Summary of Fit:

- **RSquare:** The R^2 value

ANOVA Table:

- **Prob F :** Whole model p-value

- **F Ratio:** MS_{Model}/MSE

Quick Reference

Margin of Error: Half-width of CI (precision measure)

Line Plot: Scatterplot with regression line

When correlation misses relationship: Curved patterns, discrete groups